

Creating a Low Cost, 3D Printed VTOL Drone for Accessible Surveillance Solutions

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Abstract—This project presents the development of a low-cost, 3D-printed vertical take-off and landing (VTOL) drone designed for accessible aerial surveillance applications. By utilizing the Flightory Stallion design with LW-PLA filament, we aimed to optimize flight time and minimize costs. The drone was segmented into ~60 parts for assembly onto carbon fiber tube reinforcements. The system incorporates a MATEKSYS F405-VTOL flight controller, DJI FPV components, and a Raspberry Pi Zero W running custom Python software to automate image capture and terrain mapping. Although full assembly was hindered by supply issues, the project demonstrates a viable platform for future implementation and testing of an affordable surveillance drone platform.

I. INTRODUCTION

We were motivated to try and build this because we had seen many natural disasters through the world across our diverse backgrounds and we thought that we could benefit a lot of people by making something like this. Whether it is in the aftermath of the hurricanes in the American South, or the earthquakes in West China, we believe that we can make a difference through our work.

The problem that we are attempting to solve with this project is that while surveillance drones, specifically for search and rescue efforts have proven to be extremely effective, they come with a hefty cost that cannot be fronted by low income areas. We hope to solve this by using low cost, but high quality materials and components that are easily accessible worldwide.

II. MATERIAL CHOICE

In order to keep our drone low cost, we opted to use a premade 3d printed fuselage design, the Flightory Stallion VTOL V1. We would have opted for the newer design but at the time of planning V1 was the newest model. In order to print this we had to learn about a new specialized pre-foamed polylactic acid (PLA) filament variation called lightweight polylactic acid (LW-PLA). The benefits of using LW-PLA over PLA is that it is 30% less dense than regular PLA filament while maintaining the structural properties of its heavier counterpart. This makes it so we can have longer flight times with our drone. The only downside of LW-PLA is that it is harder to work with because of very specific custom print settings that reduce print speed. For segments of the fuselage that would be under high stress during flight, we opted to use a more dense acrylonitrile butadiene styrene (ABS) filament. This was so we would not have to constantly replace these parts and so they did not fail during flight and break the entire drone.

III. SEGMENTING

In an attempt to distribute the weight for efficient flight and to be sure that the drone was able to withstand the demands of extended flights, we segmented the drone into 60 pieces, which effectively was 29 parts mirrored over the cross section of the drone with the nose and camera mount in addition. Segments under low stress were printed in LW-PLA with a 3% gyroid infill for maximum balance between weight reduction and strength. Segments that were under the most stress were connecting segments, such as the panel where the wings bolted into the fuselage or where the tail was glued into the tail receiver of the fuselage. These again were printed in ABS, along with any part that had to have threaded inserts for metric screws.

IV. HARDWARE

As of now, it is not believed to be an effective use of materials to build a 100% 3D printed drone, so in accordance with *Flightory's* design, we had to add two hollow carbon fiber tubes, one to support the wingspan of the drone and the other to make up the main body of the tail. To secure all of the segments together, we used CA glue (superglue) and small metric screws so that parts can be easily interchanged if needed.

V. ELECTRONICS

For the flight controller we used a **F405-VTOL** from *MATEKSYS*. This was a stacked component that also acted as the power distribution board for the rest of

the drone. For optimal motor performance we used three **Lumenier 51As** with 32 bit telemetry support. We also used three **T-Motor F90 1300KV motors**. We used a medium sized 4S battery with a theoretical flight time of 3 hours with gliding (without surveillance). The key components that make our drone different is the addition of a **DJI O4 Air Unit** and the **DJI N3 Goggles** for the ability to fly the drone remotely through first person view. We also have an onboard **Raspberry Pi Zero W** that runs software to take pictures from another onboard camera mounted to the bottom of the fuselage.

VI. SOFTWARE

For software, we are implementing a simple python program that runs on the onboard Raspberry Pi that takes the flight speed and elevation from the flight controller, and computes the intervals at which to take pictures of the ground in order to have full coverage of the target area. This will eventually have to incorporate elevation as well to have a truly full picture of the target area. The program will then splice these images into one big image that can be extracted from the drone upon landing through WiFi connection to the Pi.

VII. FUTURE WORK

We were very hopeful with this project, and dare I say that we took on more than we could handle in the pursuit of knowledge and experience. We had some

difficulties with drone assembly due to shortages and tariffs on Chinese manufactured goods, and many of our parts such as the carbon fiber tubes, or servo motors never arrived. We did the best attempt that we could, but without those critical pieces and being stuck to the confines of VMI without the ability to leave we could not complete the full assembly of the drone. As far as the software goes, we have a generally sound system for taking the images and mapping them, but it is not flight tested. In the future, we would like to make the drone airworthy, and fully test out the code that we have written. In addition, we would also like to try to see if we can extend the range of the connection so that we can see the map generated in real time from the ground station. Further problems were created by low print quality initially when we first started printing the segments out of LW-PLA. We had never used this material before and only one of us had experience with 3D printing, so it took time that we did not have the luxury for to hone in on the proper print settings.

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